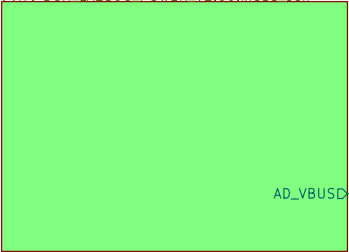
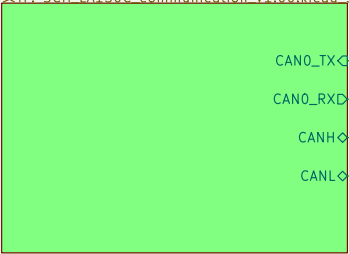


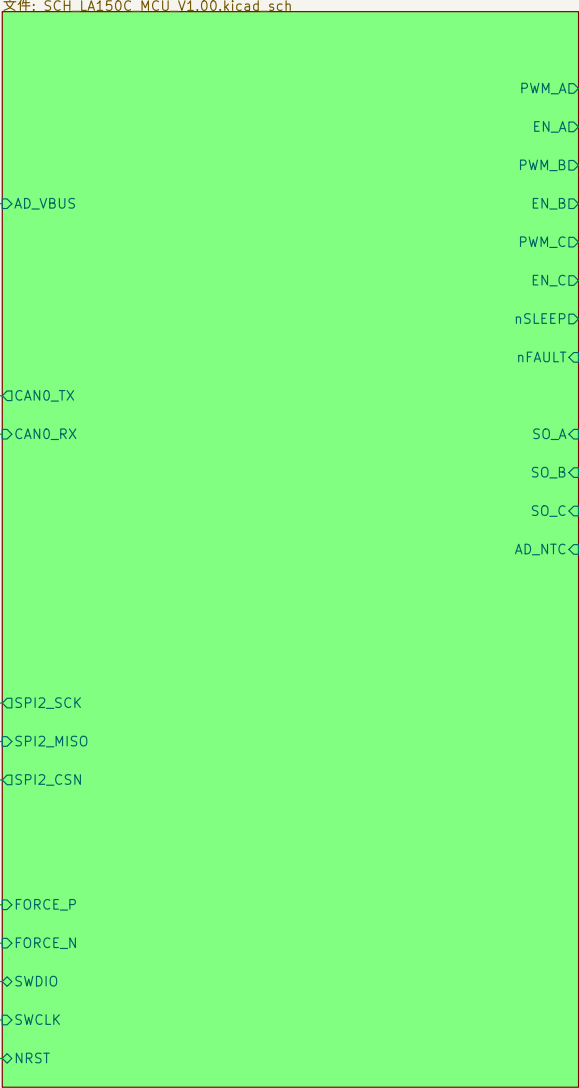
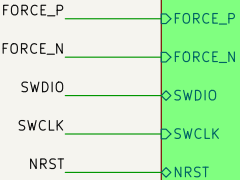
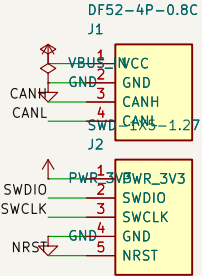
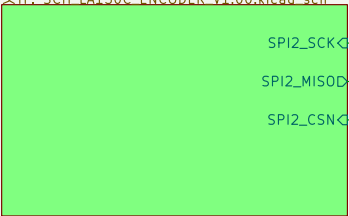
LA150C RDIVER V1.00 – TOP
层次页: POWER / MCU / MOTOR / ENCODER / COMMUNICATION
电源流向: VBUS_IN(8-14V) → [P-MOS 反接保护] → VBUS → [SCT2230MLUAR 同步降压] → PWR_3V3 → [MP6543H V3P3 + 磁珠] → PWR_3V3A
微控制器接口: DF52-4P-0.8C PIN1 VCC / PIN2 GND / PIN3 CANFD_H / PIN4 CANFD_L (线序与 Q502 规格书一致); U/V/W 与 SWD 不引出到该连接器
文件: SCH_LA150C_POWER_V1.00.kicad_sch



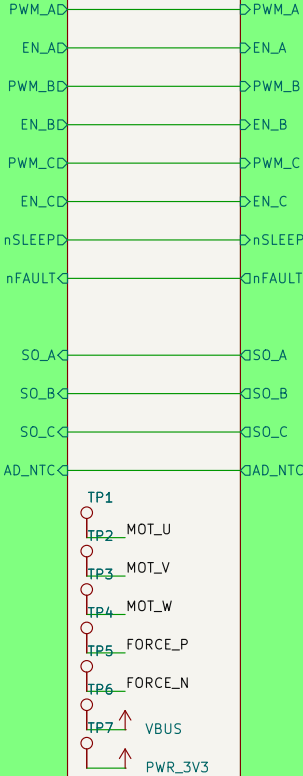
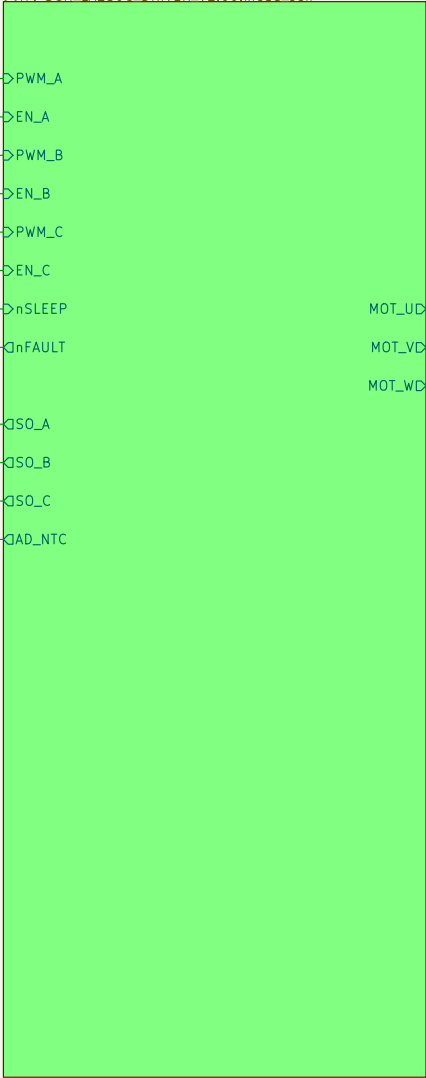
communication
文件: SCH_LA150C_communication_V1.00.kicad_sch



ENCODER
文件: SCH_LA150C_ENCODER_V1.00.kicad_sch



MOTOR
文件: SCH_LA150C_DRIVER_V1.00.kicad_sch



测试点: TP1/2/3 = MOT_U/V/W; TP4/5 = FORCE_P/N (一维力传感器预留)
TP6 = VBUS 8-14V; TP7 = PWR_3V3 3.3V ±5%

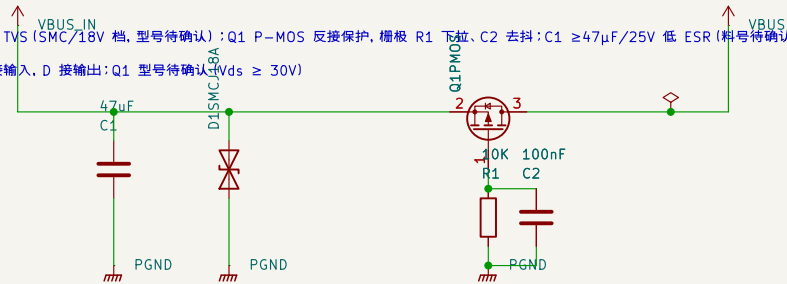
Rev	Date	Author	Changes
V1.00	2026-10-04	Codex	Altium 导入 → 脚本化重绘

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LA150C RDIVER V1.00 – POWER
母线 VBUS_IN(8-14V) → 反接保护 → VBUS → SCT2230MLUAR 同步降压 → PWR_3V3

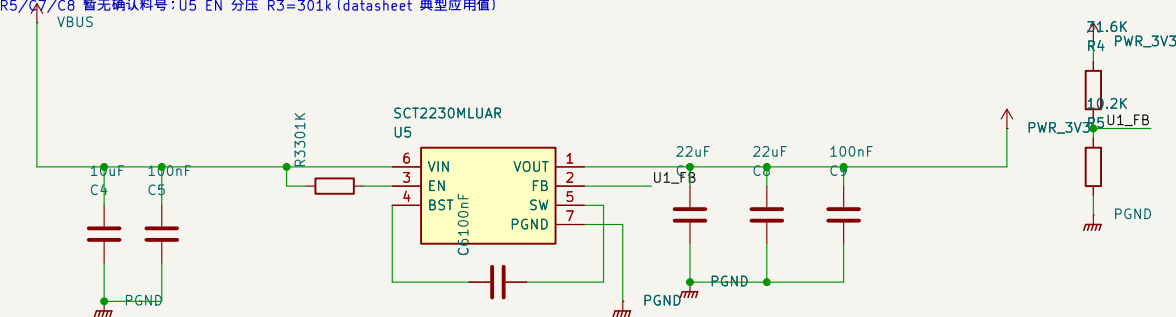
A 输入保护与母线电容

D1 母线 TVS (SMC/18V 档, 型号待确认) ; Q1 P-MOS 反接保护, 栅极 R1 下拉, C2 去抖; C1 $\geq 47\mu\text{F}/25\text{V}$ 低 ESR (料号待确认)
Q1: S 接输入, D 接输出; Q1 型号待确认 ($V_{ds} \geq 30\text{V}$)



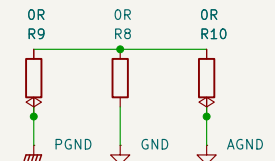
B 同步降压 3.3V – U5 SCT2230MLUAR

U5 按 datasheet Figure 8 (12V→3.3V/2A) 配置: R3 301k 使能分压、C6 100nF BST-SW、R4 31.6k / R5 10.2k 反馈、C7/C8 $2 \times 22\mu\text{F}$ + C9 100nF 输出
R3/R4/R5/C7/C8 暂无确认料号; U5 EN 分压 R3=301k (datasheet 典型应用值)



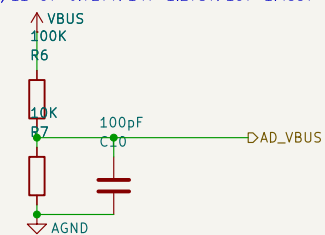
D 地网络星形连接

PGND / AGND / GND 各经 0Ω 在单点汇合
(芯片地星形连接, 避免大电流回流污染采样参考)



C 母线电压检测 → AD_VBUS

分压比 $1/11: 8\text{V} \rightarrow 0.727\text{V}, 14\text{V} \rightarrow 1.273\text{V}, 16\text{V} \rightarrow 1.455\text{V}$; RC 由 $100\text{k} \parallel 10\text{k} \times 100\text{pF} \approx 0.91\mu\text{s}$



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LA150C 集成驱动器原理图

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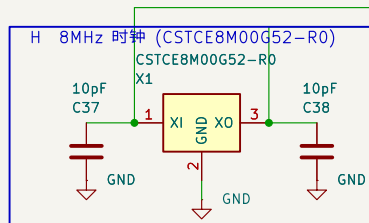
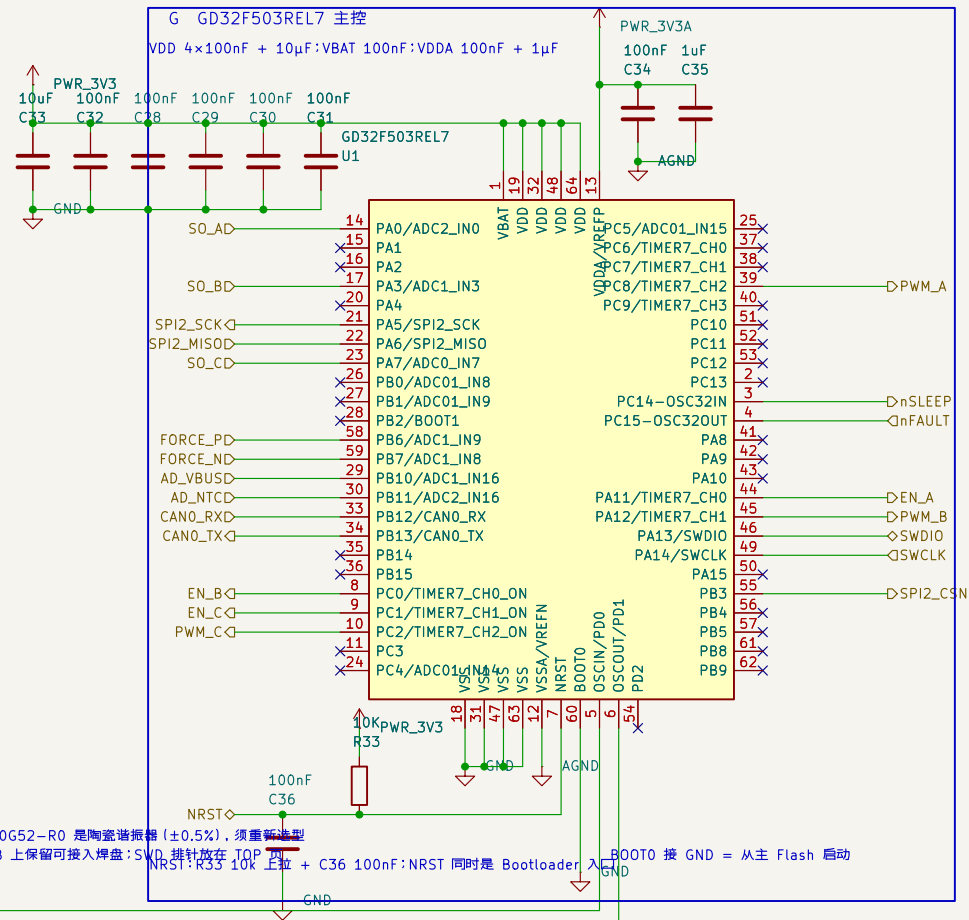
Date

2026-10-04

Rev

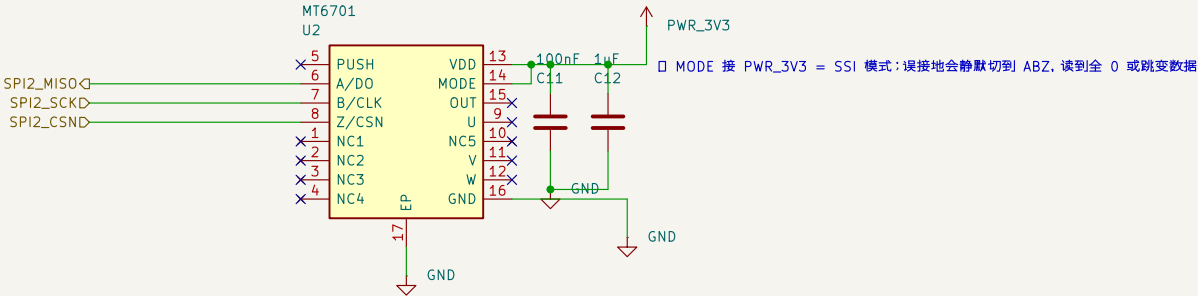
V1.00

A4



LA150C RDIVER V1.00 – ENCODER
MT6701 磁编码器 SSI:14bit 角度 + 4bit 磁场状态 + 6bit CRC, CLK 8MHz

E 磁编码器 MT6701 (SSI)
A/D0: SSI 数据输出, CLK 上升沿驱动



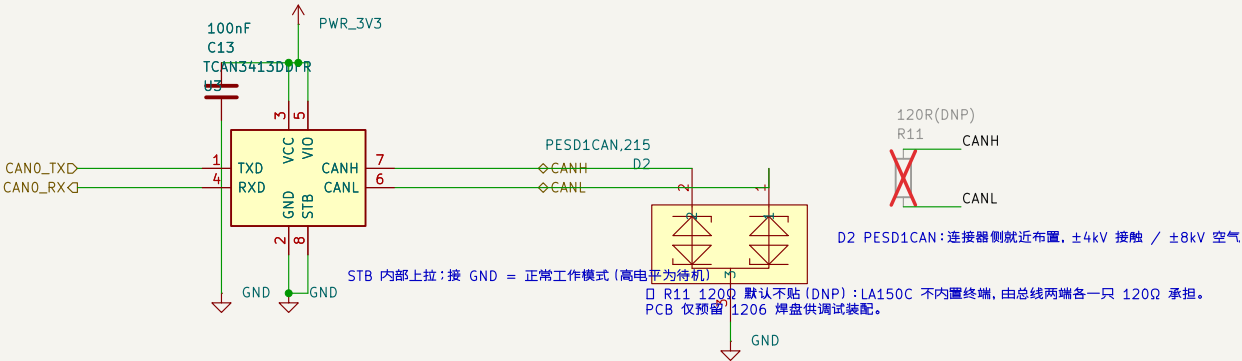
SSI:CSN 拉低 → 24 个 CLK → 14bit 角度 + 4bit 磁场状态 + 6bit CRC(X^6+X+1 , MSB first)。帧长 $3\mu s + 5\mu s$ 传播延迟 → 125kHz 更新率, 远高于 FOC 电流环需求。

磁体 $\varnothing 6 \times 2.5\text{mm}$ N35SH 径向两极, 气隙 0.5-2.0mm (典型 1.0mm), 偏心 $\leq 0.3\text{mm}$ (结构件, 不上图)

VDD 去耦 $100\text{nF} + 1\mu\text{F}$:U/V/W 与 PUSH/OUT/NC 在 SSI 模式下不用, 全部置 no_connect

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F CAN FD 收发器 TCAN3413DDFR



LA150C 仅保留 CAN FD + CANopen CIA402, 不保留 RS485 (PB6/PB7 已改作一维力传感器预留输入)。

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